

## Assignment Week – 7 (Full mark 15)

Q1. Identify the true and statements

1 x 4 = 4

- I. The value of activity coefficient ( $\gamma_2$ ) for the same type of electrolytes follows the same trend.  
(a) T (b) F
- II. The value of  $I_t$  for the mixture of 2 independent on the nature of the 2 or more electrolytes is individual ions.  
(a) T (b) F
- III.  $\gamma_{\pm}$  of a given electrolytes is proportional to the square root of the charge on the ions.  
(a) T (b) F
- IV. The Value of  $I_t$  for a in a mixture is 4 times less than that done by  $\text{Na}^+$  &  $\text{K}^+$ .  
(a) T (b) F

Q2. Which one is correct about the speed of group I ions.

2

- a)  $\text{Li}^+ > \text{Na}^+ > \text{K}^+ > \text{Rb}^+ > \text{Cs}^+$   
b)  $\text{Li}^+ \sim \text{Na}^+ > \text{K}^+ > \text{Rb}^+ > \text{Cs}^+$   
c)  $\text{Li}^+ < \text{Na}^+ < \text{K}^+ < \text{Rb}^+ < \text{Cs}^+$   
d) None

Q3. Grotthuss mechanism is applicable only for-

1

- a)  $\text{Na}^+$  ion  
b)  $\text{Cs}^+$  ion

c)  $H^+$  ion

d)  $K^+$  ion

Q4. Debye-Hückel-Onsager proposed which relation to measure equivalent conductance ( $\lambda$ ) for strong electrolyte in water.

2

a)  $\lambda = \lambda_0 - [A + B\lambda_0]\sqrt{C}$

b)  $\lambda_c = \lambda_0 - b\sqrt{C}$

c)  $\lambda = \lambda_0 + [A + B\lambda_0]\sqrt{C}$

d)  $\lambda = \lambda_0 + b\sqrt{C}$

Q5. If the Specific conductance of a saturated Solution of  $SrSO_4$  is  $1.5 \times 10^{-4}$  and that of pure water is  $1.5 \times 10^{-6} \text{ mho cm}^{-1}$  at  $25^\circ\text{C}$ , calculate solubility of  $SrSO_4$  in g/litre unit. Given;  $l_c^0 = 59.2$  and  $l_a^0 = 79.8$ , Molecular weight of  $SrSO_4 = 92.8$

2

a) 0.050

b) 0.080

c) 0.099

d) 0.060

Q6. The dissociation Constant for a weak acid Can be determined by using –

2

a)  $K_a = K'_a \cdot f_{\pm}^2$

b)  $K_w = C_{H^+} \cdot C_{OH^-}$

c)  $K_2 = \frac{(K_1 + C)}{K_1 C} \cdot C_{H^+}^2$

d) both a & b

Q7. The equivalent conductance of a 0.02 N solution of  $\text{CH}_3\text{COOH}$  is  $15.2 \text{ mho cm}^2 \text{ eq}^{-1}$ .

The ionic conductances at infinite dilution are:

$$\lambda_{\infty}(\text{H}^+) = 350.0$$

$$\lambda_{\infty}(\text{CH}_3\text{COO}^-) = 40.0$$

Calculate the degree of dissociation ( $\alpha$ ) and the dissociation constant ( $K_a$ ).

a)  $0.039, 6.32 \times 10^{-5}$

b)  $0.76, 3.16 \times 10^{-4}$

c)  $0.039, 3.16 \times 10^{-5}$

d)  $0.76, 6.32 \times 10^{-5}$

$$\lambda_{\infty} = \lambda_{\infty}^+ + \lambda_{\infty}^- = 350.0 + 40.0 = 390.0 \text{ mho cm}^2 \text{ eq}^{-1}$$

$$\alpha = \frac{\lambda_c}{\lambda_{\infty}} = \frac{15.2}{390} \quad \alpha = 0.039 \quad (2)$$

for weak electrolyte  
 $K_a = \frac{\alpha^2 c}{1 - \alpha}$  since  $\alpha$  is small

Q8. The speed with which an ion moves under a unit potential gradient (1 V/cm) is called:

a) Specific conductivity

b) Ionic mobility

c) Molar conductance

d) Transport number

$$K_a = \alpha^2 c$$

$$K_a = (0.039)^2 \times 0.02$$

$$K_a = 3.042 \times 10^{-5}$$

Q9. Kohlrausch's law states that:

a) At infinite dilution, the molar conductivity of an electrolyte equals the sum of the ionic conductances of its ions.

b) At moderate dilution, the molar conductivity of an electrolyte equals the sum of ionic conductances of its ions.

c) Valid at all concentrations.

d) None of the above.

(2)

**Q10.** The equivalent conductance at infinite dilution for NaOBu, NaCl and HCl are 83, 127 and 426 mho cm<sup>2</sup>. What is the equivalent conductance at infinite dilution of BuOH?

- a. 470 mho cm<sup>2</sup>
- b. 382 mho cm<sup>2</sup>
- c. 299 mho cm<sup>2</sup>
- d. 343 mho cm<sup>2</sup>

$$\lambda_{\text{BuOH}} = \lambda_{\text{HCl}} + \lambda_{\text{NaOBu}} - \lambda_{\text{NaCl}}$$

$$\lambda_{\text{BuOH}} = 426 + 83 - 127$$

$$382 \text{ mho cm}^2 \quad (2)$$